

Statistical supplement — VALIDATED (cite from this document only)

_Every value below carries a receipt (F-number → receipts/) and an offline recompute script (scripts/). Corrections from the 2026-06-11 campaign are applied; superseded values appear only in CLAIMS_LEDGER.md. Consolidation: strict both-orders; win-rate counts ties as ½; pooled frontier statistics are question-clustered._

V1. Within-model lever effects (answerable-only; Y = survives global Holm, family=36) — [F01]

Retrieval (RAG over base) — robust everywhere

corpus	llama8b	phi4mini	minicpm1b
base (n=293)	84.1% Y	84.8% Y	81.2% Y
COVID (n=400; minicpm 393–395)	86.4% Y	83.6% Y	79.2% Y
QASPER (n=302)	59.6% Y	60.9% Y	62.3% Y

Fine-tuning (LoRA over base) — small; 7/9 Y: base 48.8% ns / 55.8% Y / 53.2% Y; COVID 53.0/54.4/54.9 all Y; QASPER 53.5 Y / 51.5 ns / 52.9 Y. Near-nulls TOST-equivalent (±0.05).

RAFT (lora_rag over rag) — 2/9 Y: COVID-MiniCPM 58.0% (p=3.7e-6), QASPER-Llama 54.6% (p=7.6e-4).

Global correction: Holm 27/36 (retrieval 18/18, LoRA 7/9, RAFT 2/9); BH 29/36 (8/9, 3/9).

Note: "RAG-over-LoRA" = lora_rag vs lora (retrieval added on top of LoRA) — footnote this label.

V2. Lever × corpus interactions — [F02]

effect	base vs COVID	base vs QASPER	COVID vs QASPER
RAG-over-base	G=2.49, p=0.115	G=26.48, p=2.7e-7	G=17.64, p=2.7e-5
RAG-over-LoRA	G=0.30, p=0.584	G=39.80, p=2.8e-10	G=38.73, p=4.9e-10
LoRA-over-base / RAFT-over-RAG	all ns (min p=0.091 / 0.100)		

Robustness: question-block permutation p=5.0e-4 (0/2000) for the headline interaction.

Scope: per-model, the base-vs-QASPER retrieval interaction holds for llama8b (G=25.5) and phi4mini (G=4.4) but not minicpm1b (G=0.67, saturated) — state non-uniformity.

V3. Mechanism — decomposition & homogeneity (CORRECTED rows) — [F03]

Identity recall·hit-win + (1–recall)·miss-win = aggregate holds exactly on all corpora.

corpus (llama8b, answerable)	recall@6	hit-win (n)	miss-win	aggregate	TF-IDF centroid	pairwise cos
base (n=293)	0.294	0.878 (86)	0.826	0.841	0.235	0.053

corpus (llama8b, answerable)	recall@6	hit-win (n)	miss-win	aggregate	TF-IDF centroid	pairwise cos
COVID (n=400)	0.813	0.931	0.573	0.864	0.136	0.018
QASPER (n=302)	0.232	0.807	0.532	0.596	0.140	0.019

Controlled contrast: base vs QASPER sit in a similar low-recall band (.294 vs .232) while the homogeneity gap is **1.68×** (centroid) / **2.8×** (pairwise) and miss-win differs .826 vs .532; the ~33× size gap is not the driver. (COVID/QASPER rows corrected from stale n=150/112 artifacts.)

V4. OOC honesty (upgraded) — [F04, F12]

Win-rate endpoints (RAG-over-base on unanswerables, pooled 3 models): base **0.705** (n=156; text-clustered 95% CI [0.59, 0.83]; 27 unique texts) vs QASPER **0.221** (n=294; CI [0.19, 0.26]). Holds per-model: base .73/.61/.78, QASPER .27/.15/.24.

Behavioral mechanism (scripted classifier; llama8b):

corpus	correct-abstain on OOC, base → RAG	false-abstain on answerable, base → RAG
base (homogeneous)	61.5% → 100%	85.7% → 11.3%
QASPER (heterogeneous)	63.3% → 17.3%	—

Risk-coverage on OOC (every engaged answer is an error): QASPER-RAG coverage 0.939 (self-rejects 6%); base-RAG 0.058. Engaged-answer quality gain from RAG on QASPER: +0.007 ROUGE-L (negligible) — the harm is behavioral (engagement), and its sign flips with corpus homogeneity. Retired: the 36/98-vs-42/98 spot-check (direction was backwards). ECE infeasible (no logprobs).

V5. Frontier closed-book (access gap) — [F05]

	local not-lose (panel / single)	outright frontier loss (panel)
8B+RAG	95.8% / 98.8%	75.0%
1B+RAG	91.4% / 96.0%	65.9%

Failure modes (scripted classifier, rules in receipt): GPT attempts 49.9% (0 formal abstentions); Claude pure-abstains 76.8%, DeepSeek 65.8% (threshold-sensitive band ≈66–77%). Pure abstention on OOC is never scored as a loss (0/98).

V6. Frontier same-retrieval, clean-judged (THE HEADLINE) — [F06, F17]

comparison	net win-rate [95% CI, q-clustered]	local not-lose	tie	q-sign test
vs 8B — base all (n_q=344)	0.600 [0.580, 0.620]	0.787	77.4%	98:10, p=2.7e-19
vs 8B — base answerable	0.618 [0.595, 0.641]	0.749	73.3%	—

comparison	net win-rate [95% CI, q-clustered]	local not-lose	tie	q-sign test
vs 8B — QASPER answerable (n_q=149)	0.573 [0.543, 0.603]	0.810	76.5%	44:12, p=2.1e-5
vs 1B — base all	0.720 [0.695, 0.746]	0.546	53.2%	175:7, p=4.0e-43
vs 1B — QASPER answerable	0.764 [0.726, 0.802]	0.441	40.9%	108:6, p=2.7e-25

Local decisive wins: 14/1030 (1.4%). Frontier wins 91–97% of decisive pairs (per-cell sign tests $p \leq 7e-3$... 2.4e-44; Wilson CIs in receipt). Capped-panel cross-checks: pooled 0.573/0.774; gpt+claude pool 0.591/0.752; q-clustered capped sign test 60:26 $p=3.2e-4$.

Robustness chain: judging regime (capped panel 0.573 → clean 0.600, $\Delta+0.027$) · aggregation rule (net 0.540–0.575 over 5 panel variants; strict ≥ 2 -decisive raises not-lose to 0.835) · retrieval success (V7) · corpus (base $\approx$ QASPER).

V7. Hit/miss stratification — tie mass is capability parity (NEW) — [F11]

HIT = gold source chunk in the shared top-6.

pooled cell	HIT net win-rate	MISS net win-rate	HIT–MISS Δ [95% CI]
base vs 8B	0.607 [0.568, 0.649] (86 q)	0.598	+0.009 [−0.037, +0.057]
base vs 1B	—	—	+0.022 [−0.038, +0.082]
QASPER vs 8B	0.616 (36 q)	0.559	+0.057 [−0.029, +0.142]
QASPER vs 1B	—	—	+0.037 [−0.051, +0.124]

All deltas ≈ 0 : the frontier edge does not grow when retrieval succeeds; the 8B ties the frontier ~77% even with the gold chunk in context. Mutual-blindness hypothesis rejected.

V8. QASPER arm — replication & contamination control — [F07]

Closed-book stratification (the inversion): answerable fwr **0.313** / not-lose 0.982 (frontier loses, same as private corpus) vs OOC fwr 0.833 (abstention correct); abstention .669/.771 by stratum. RAG parity: pooled 0.535, answerable 0.524 / not-lose 0.906. Heterogeneity-premium NULL: QASPER $0.573 \leq$ base 0.600–0.618 under clean judging. 1B more competitive on QASPER (0.572).

V9. DeepSeek v2 (token-cap fix) — [F08]

v2 cells: single 0.5609 / 0.6754; panel 0.5536 / 0.7058 (n=345). Deltas vs v1: +0.011–0.036. Answer-level proof: 16/24 v1 decisive local wins dissolved; 14 are exact-512-token truncations; v1 had 41 exact-512 answers; v2 has 0 cap hits / 0 CoT leaks. Attrition immaterial (0 dropped).

V10. Arena, gold anchor, ROUGE — [F09]

GOLD dominates all 6 BT rankings (base-rag GOLD 56.5 vs best 0.61). Caveat: 3 never-beaten GOLD anchors (base-lora_rag 192.8; qasper-rag 275.0; qasper-lora_rag 301.1) are non-convergent MLEs — report as lower bounds at n_iter=400. Best model is corpus-dependent (COVID→Llama, QASPER→Phi, base flips with config). Panel α 0.631/0.641/0.566. ROUGE-L grid reproduces exactly (base-llama 0.223/0.480/0.172/0.559); rag,lora_rag > base,lora on all 9 rows; repetition-collapse is load-bearing for MiniCPM rows.

V11. Confounds — [F10, F13, F15, F17]

item	value	verdict
Position bias P(first wins)	.490 / .4965 / .4953	clean
Verbosity P(longer wins)	.5009 / .4286 / .4842	clean (COVID favors shorter)
Corpus routing / empty retrievals	1.0 all 18 cells / 0	clean
Completeness	400/config (base 345), 0 empty; COVID-minicpm 393–395 (76 missing rows)	clean / disclose n
MiniCPM truncation (QASPER)	lora_rag 136, lora 78 , rag 23 /400; phi COVID 2; llama 0	disclose (incl. lora)
Tie rates	.5273 / .5255 / .7809	QASPER low power → n=400 + pooling
Eval hash-locks	SHA-256 match across all run snapshots	clean
Eval dedup	base 315/345 unique (OOC 27/52); QASPER 381/400 ; COVID 398/400	disclose all three
Corpus	28 published papers (13 OpenAlex W-ids), 593 chunks, 416,698 tokens ; possible dup: "Planning Early" ×2	provenance stated; diff dup
Judge panel	α .565 (frontier); mistral primacy 73.1% first-position (contained by both-orders); kimi flip 9.4%; rule-sensitivity net 0.540–0.575	disclose; design works
Evidence-window artifact	tie→frontier 20:0 (p=1.9e-6); 14–15/20 verbosity-driven → fixed by full-evidence + de-verbosity judging	corrected by design
Frontier prompt symmetry	byte-identical system prompt + retrieval (GPT token-exact, delta=11 const)	clean
Decoding	max_tokens=512 (v1 gen), silent temp-drop-on-400 fallback exists	disclose

V12. Cost (corrected) — [F14]

run	nominal logged	honest
v1 frontier pilot	\$0.95	≈\$14.7 (gen \$9.73 + judge ≈\$5.0)
panel / deepseek-v2 / evidence / qasper-arm / clean-rejudge	6.69 / 3.45 / 0.67 / 5.17 / 14.41	as logged (correct pricing)
project total (API)	~\$31.3 nominal	≈\$45 honest

Per-1k-questions frontier API (gen only): closed-book gpt \$1.44 / claude \$1.70 / deepseek \$0.29; same-RAG (~4.5k-tok context) gpt \$6.31 / claude \$16.34 / deepseek \$2.20. Local 8B: \$0 marginal API cost (hardware assumption disclosed; latency and TCO not measured — make no speed claims).